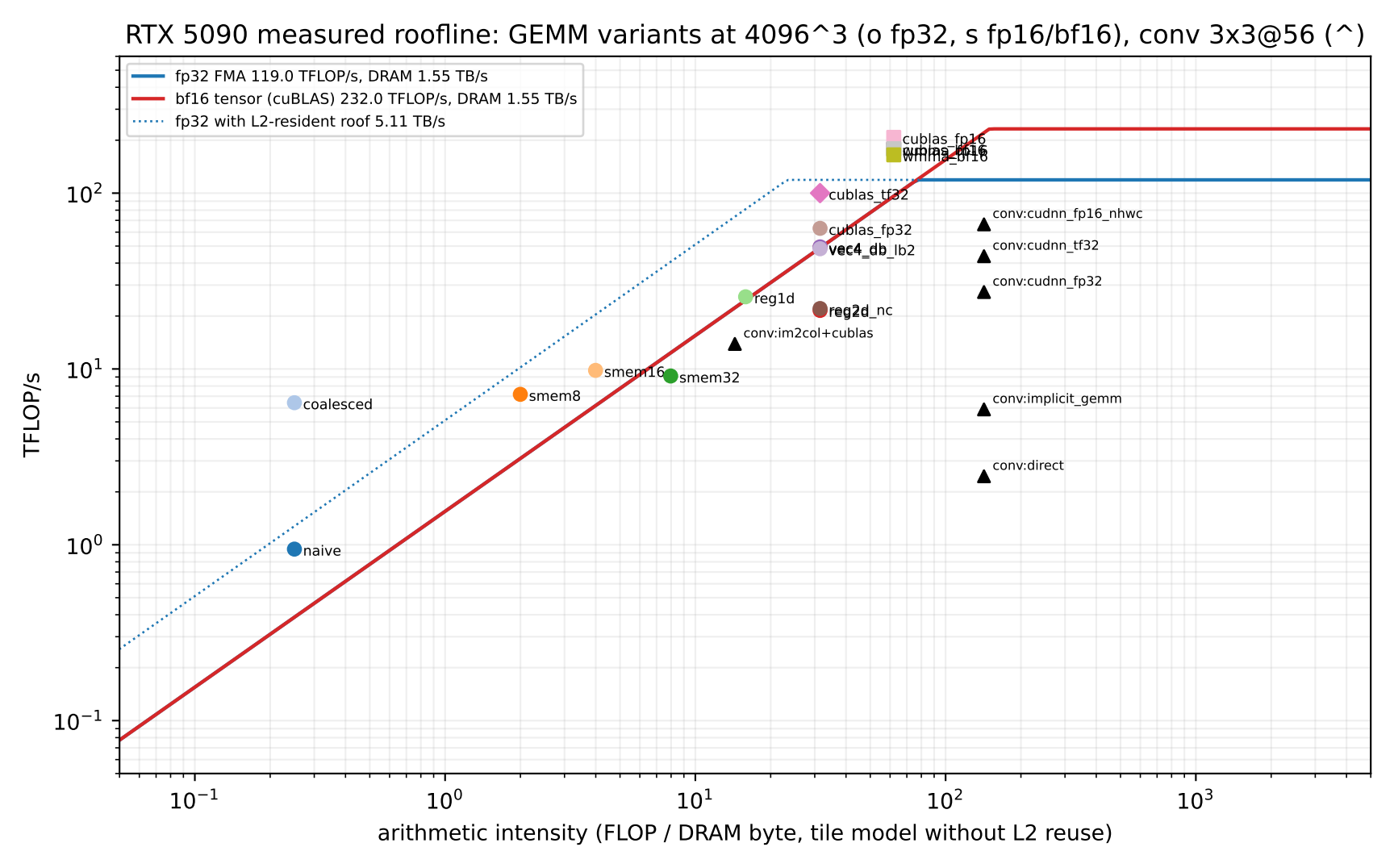
RTX 5090 measured roofline: GEMM variants at 4096^3 (o fp32, s fp16/bf16), conv 3x3@56 (^)
fp32 FMA 119.0 TFLOP/s, DRAM 1.55 TB/s
bf16 tensor (cuBLAS) 232.0 TFLOP/s, DRAM 1.55 TB/s
fp32 with L2-resident roof 5.11 TB/s
TFLOP/s
arithmetic intensity (FLOP / DRAM byte, tile model without L2 reuse)
cublas_fp16
cublas_bf16
wmma_bf16
cublas_tf32
cublas_fp32
vec4_db_lb2
reg1d
reg2d_nc
conv:im2col+cublas
smem16
smem32
smem8
coalesced
naive
conv:cudnn_fp16_nhwc
conv:cudnn_tf32
conv:cudnn_fp32
conv:implicit_gemm
conv:direct